Label Compression and Gradient Transmission: A Dynamic Account of Theory Assimilation in Scientific Machine Learning

Kevin T.N.
Independent Researcher
jkdkr2439@gmail.com

Abstract

Debates on theory-ladenness in scientific machine learning often emphasize the apparent procedural independence of model training from domain-theory: optimizers, hyperparameter search, and many architectural choices can be applied across domains without disciplinary understanding. This paper accepts a local version of that observation while arguing that it misses a distinct mechanism by which domain-theoretic commitments can nevertheless shape learned models over time. In supervised scientific machine learning, labels and evaluation rules commonly operationalize disciplinary ontologies, measurement conventions, and expert criteria. Such supervisory signals are low-bandwidth outputs of high-bandwidth theory and expertise. Because learning objectives are defined relative to these signals, gradient-based optimization iteratively propagates their constraints through parameter updates. I formalize this idea as *label compression* (theory-conditioned supervision as compressed judgment), *objective-level encoding* (theory entering via the loss), and *gradient transmission* (iterative propagation of the encoded constraints). The resulting picture is dynamic rather than static: models may be procedurally theory-indifferent yet dynamically theory-assimilating. The account clarifies why parameter opacity undermines traceability without establishing theory-independence, and it sharpens normative questions about when explicit domain-theory should be integrated into target definitions, labeling practices, and objective design to secure robustness, validity, explanation, and causal generalization.

Keywords: theory-ladenness; scientific machine learning; supervision; labels; objective functions; gradient descent; dataset shift; robustness; operationalization.

1 Introduction

Scientific machine learning (ML) has intensified long-standing philosophical disputes about how theory structures inquiry. A common intuition in philosophy of science holds that scientific activity is theory-laden: theoretical commitments shape what is measured, how evidence is produced, which models are built, and how results are interpreted. Yet contemporary ML workflows often appear to complicate this picture. Optimizers such as stochastic gradient descent (SGD), standard architectural templates, and automated hyperparameter search can be applied across scientific domains without the modeler possessing domain expertise. This motivates a claim of *procedural theory-indifference* in model construction: in many contexts, one can train high-performing models while remaining largely agnostic about the scientific theory of the target domain.

Termine, Ratti, and Facchini (TRF) develop this point in the context of the theory-ladenness debate by distinguishing model *construction* from model *use*, and by arguing that construction can be significantly theory-indifferent even when scientific use remains theory-laden [1]. They also introduce *theory-infection* to capture a weaker mode of theory-ladenness: trained models may passively inherit priors embedded upstream in curated datasets.

This paper preserves the spirit of TRF's distinction while proposing a different explanatory target. Rather than asking only whether a construction step is theory-indifferent, I ask: *by what mechanism can theory shape a trained model over time even when training procedures are domain-agnostic?* The central thesis is dynamic:

> **Dynamic theory assimilation thesis.** In supervised scientific ML, domain-theoretic commitments are often compressed into supervisory signals (labels and evaluation rules), encoded into the objective function, and iteratively transmitted into model parameters through gradient-based optimization.

This thesis does not claim that models "understand" theory. It claims a weaker, structurally grounded point: models can align with constraints implicit in theory-conditioned supervision. The upshot is a three-layer picture—procedures, objectives, and dynamics—that clarifies how procedural theory-indifference can coexist with theory-conditioned learning trajectories.

Scope

The core argument is developed for supervised learning in scientific contexts. Extensions to self-supervised learning and reinforcement learning are briefly discussed in Section 10. The paper is conceptual and formal rather than empirical; its aim is to make epistemic claims well-posed by specifying where theory enters the pipeline and how it can propagate.

2 Three Layers: Procedure, Objective, and Dynamics

The phrase "theory-ladenness" can conflate distinct questions. To reduce ambiguity, I distinguish three analytic layers.

Definition 1 (Procedural layer)**.** *The* procedural layer *concerns the algorithmic operations used to fit a model: choice of optimizer, update rules, generic architectural patterns, and routine training heuristics.*

Definition 2 (Objective layer)**.** *The* objective layer *concerns how targets, labels, and evaluation criteria are specified: operational definitions of phenomena, labeling rules, measurement conventions, and loss functions.*

Definition 3 (Dynamic layer)**.** *The* dynamic layer *concerns how repeated optimization updates reshape parameters over time in response to the objective: learning trajectories, convergence behavior, and the progressive alignment induced by iterative error minimization.*

A key point is that these layers can dissociate. An optimizer can be domain-agnostic (procedural theory-indifference) while the objective is deeply theory-conditioned (objective-level encoding). Dynamics then propagate the objective's constraints into the learned representation (gradient transmission).

3 Preliminaries: Supervised Learning Setup

Let (x, y) denote an input-label pair, where $x \in \mathcal{X}$ and $y \in \mathcal{Y}$. A parametric model $f_\theta : \mathcal{X} \to \mathcal{Y}$ is trained by minimizing an empirical risk:

$$\min_\theta \ \frac{1}{n} \sum_{i=1}^{n} \ell(f_\theta(x_i), y_i)\,, \tag{1}$$

possibly with regularization. Gradient-based optimization updates parameters by

$$\theta_{t+1} = \theta_t - \eta \nabla_\theta L(\theta_t), \tag{2}$$

where L is the (possibly stochastic) training objective and η is a step size [2].

Nothing in (2) requires domain-specific understanding. This supports procedural theory-indifference. But the objective L is defined *relative to* y and to evaluation conventions. In scientific contexts, those conventions are frequently theory-conditioned. The remainder of the paper explicates this conditioning and its dynamic consequences.

4 Label Compression

Supervision looks informationally simple: a label can be binary, categorical, or a scalar. Yet in scientific practice, labels often summarize complex theoretical and institutional work.

Definition 4 (Operationalization)**.** Operationalization *is the process by which an intended scientific phenomenon is mapped into measurable proxies, labeling criteria, and evaluation standards, often using domain-specific ontologies and measurement conventions.*

In medical imaging, for example, a label such as "malignant" is not an unstructured fact. It reflects diagnostic protocols, measurement standards, threshold choices, and often tacit expertise. Similar points hold for labels in genomics, remote sensing, materials science, and the social sciences where categories are theory- and institution-shaped.

Proposition 1 (Label compression)**.** *In supervised scientific ML, labels and evaluation rules often function as compressed outputs of domain-conditioned operationalization. They are low-bandwidth supervisory signals that presuppose high-bandwidth disciplinary commitments (ontologies, measurement conventions, thresholds, and expert criteria).*

Proposition 1 is not the claim that labels contain the whole of a domain-theory. Rather, labels typically compress those parts of disciplinary practice that matter for adjudicating correctness under a specified task. This compression can be faithful or distorting.

4.1 Faithful vs. distorting compression

Compression can preserve the invariants that matter for a task, or it can encode biases, artifacts, and brittle proxies. In ML system design, this is the difference between a label that tracks a stable phenomenon and a label that tracks a contingent measurement artifact. The normative stakes of supervision depend on which kind of compression one has implemented [5, 6].

5 Objective-Level Theory Encoding

If labels are theory-conditioned compressions, then the learning objective is theory-conditioned in a precise sense: the loss is defined relative to those labels.

Proposition 2 (Objective-level encoding)**.** *When supervision (x, y) is theory-conditioned via operationalization, the learning objective $L(\theta)$ is implicitly theory-conditioned: the loss specifies deviations from a compressed theoretical judgment (the label and associated evaluation rules), and thereby defines what counts as error, relevance, and success.*

Two clarifications matter.

1. **Encoding is operational, not symbolic.** The objective does not contain a sentence-level representation of theory. It contains a metricized constraint derived from theory-conditioned judgments.

2. **Conditioning can be upstream.** Even if the modeler does not consult domain theory, the objective may still be shaped by the domain experts who produced the labels and the evaluation conventions.

This reframes a common misunderstanding. Parameter-level opacity (lack of human-interpretable semantics for weights) may limit traceability of theoretical influence, but it does not follow that theory plays no epistemic role. The objective is already a theory-conditioned filter on what structure can be learned.

6 Gradient Transmission and Temporal Theory Assimilation

We now connect objective-level encoding to a dynamic mechanism of assimilation.

6.1 Gradient transmission

Proposition 3 (Gradient transmission)**.** *Given an objective $L(\theta)$ defined relative to theory-conditioned supervision, gradient-based optimization propagates the objective's constraints into parameter updates: each step in (2) is driven by deviations from the compressed theoretical signal and therefore transmits that signal (in constraint form) into model parameters.*

This is a structural, not semantic, claim. The model need not "represent" theory in an interpretable way. It suffices that the optimization dynamics align the model with the objective's constraints.

6.2 Temporal assimilation

Single gradient steps are negligible; assimilation is temporal. Under standard learning assumptions (finite capacity, bounded optimization, fixed objective), repeated updates drive the model toward a local optimum of a theory-conditioned risk surface.

Proposition 4 (Temporal assimilation (objective-relative))**.** *Over iterative training, models tend to approximate the structural constraints encoded in theory-conditioned supervision* relative to the objective and training regime*. This constitutes a dynamic form of theory assimilation: the model progressively aligns with invariants that the operationalization makes salient, while ignoring structure outside the task-defined objective.*

Proposition 4 is intentionally indexed to objectives and regimes. It does not imply robust generalization beyond the distribution or task; it states how constraints propagate under repeated optimization.

6.3 Opacity, traceability, and the limits of the inference to independence

A frequent move in discussions of deep learning is to infer from parameter opacity that theory does not shape the model. The present framework blocks that inference:

Opacity undermines *traceability* of influence in parameter semantics, but objective-level encoding and gradient transmission show how theory-conditioned supervision can still shape learning dynamics.

7 Failure Modes and Limits

The dynamic account clarifies not only how theory can be assimilated, but also how it can fail.

7.1 Distorting compression and spurious alignment

If labels compress distorted operationalizations, models will faithfully optimize toward those distortions. A common failure mode is spurious alignment: the model learns predictors that correlate with labels in the training regime but do not track the intended phenomenon. This is not merely "overfitting" in a statistical sense; it can be overfitting to a particular operationalization.

7.2 Annotation drift and shifting theoretical compression

Operational definitions and labeling conventions can drift over time (changes in guidelines, instruments, or ontologies). When this happens, the objective changes even if the procedure does not. The model can become misaligned because it assimilated constraints of a previous compression.

7.3 Dataset shift as operationalization shift

Distribution shift is widely recognized as a core obstacle to reliability [7, 8]. The present framework highlights a specific subtype: *operationalization shift*, where the mapping from phenomenon to label changes. In such cases, failures under shift can reflect not only statistical mismatch but also a change in what the labels mean (or which invariants they preserve).

7.4 Robustness and the role of explicit theory

Robustness and out-of-distribution generalization often require constraints that are not guaranteed by generic objectives. This motivates explicit domain-theory integration at the objective layer (e.g., invariance constraints, causal assumptions, or physically motivated inductive biases) [3, 4].

8 Relation to Theory-Indifference and "Theory-Infection"

The present account is compatible with a local thesis of procedural theory-indifference: SGD does not require disciplinary knowledge to operate. But the account adds a mechanism-level complement: even when procedures are domain-agnostic, objectives can be theory-conditioned and learning dynamics can assimilate the constraints.

TRF's *theory-infection* can be reinterpreted in operational terms: what looks like a mysterious "slipping in" of theory is often objective-level encoding via curated supervision and evaluation conventions. The emphasis shifts from locating theory in parameters (a task obstructed by opacity) to locating theory in the pipeline components that define what counts as error and success.

9 Normative Implications

The dynamic account reframes normative questions. Instead of asking whether ML "needs theory" in general, one asks: *At which layer, and for which epistemic aim, should explicit domain-theory be enforced?*

Decision grid (aim-indexed theory requirements).

- **Aim: benchmark prediction (IID).** Procedures can be domain-agnostic; domain theory is often most critical upstream for valid targets and labels.
- **Aim: robustness under shift (OOD).** Domain constraints are frequently needed at the objective layer (stress tests, invariance assumptions, meaningful evaluation).
- **Aim: explanation/mechanism.** Domain theory is typically indispensable downstream and often upstream (choice of explanatory variables and operational definitions).
- **Aim: causal intervention/generalization.** Domain theory and explicit causal assumptions become central; purely correlational objectives are rarely sufficient.

This grid also clarifies accountability: if theory is compressed into labels, then governance of labeling and evaluation becomes a primary locus of epistemic responsibility.

10 Extensions Beyond Supervised Learning

The core mechanism is clearest in supervised learning because labels explicitly anchor the objective. Two brief remarks extend the idea.

10.1 Self-supervised and contrastive objectives

In self-supervised learning, the "supervisory signal" is produced by task design (augmentations, contrastive pairs, masked prediction targets). Here, theory-conditioning can enter through the choice of pretext task and invariances it enforces. The same structural claim holds: objectives encode constraints; optimization transmits them.

10.2 Reinforcement learning

In reinforcement learning, reward functions operationalize success criteria. If reward design is domain-conditioned, then objective-level encoding and temporal assimilation arise through policy optimization analogously.

11 Conclusion

This paper proposed a dynamic account of theory assimilation in scientific machine learning. The account distinguishes procedure, objective, and dynamics; formalizes *label compression* as the compression of domain-conditioned operationalization into supervisory signals; identifies *objective-level encoding* as the entry point of theory into the loss; and articulates *gradient transmission* and *temporal assimilation* as the mechanism by which constraints propagate into parameters over time.

The result is a reconciliation of two intuitions: ML procedures can be domain-agnostic, yet learned models can still be shaped by theory-conditioned supervision. Parameter opacity undermines

traceability but does not establish independence. Finally, the normative question becomes sharper: to secure robustness, validity, explanation, and causal generalization, theory must often be addressed where it actually enters and propagates—in the operationalization of targets, labels, and objectives.

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